1.51 2D Vectors

Worked Solutions

1 a)

$$\overrightarrow{AB} = \overrightarrow{OB} - \overrightarrow{OA} = \begin{pmatrix} -3 \\ 8 \end{pmatrix} - \begin{pmatrix} 2 \\ 6 \end{pmatrix} = \begin{pmatrix} -5 \\ 2 \end{pmatrix}.$$

1 b)

$$|\overrightarrow{AB}| = \sqrt{(-5)^2 + 2^2} = \sqrt{29}.$$

1 c)

Let $\hat{\mathbf{p}}$ be the unit vector in the direction of $\overrightarrow{AB}$.

$$\hat{\mathbf{p}} = \frac{\overrightarrow{AB}}{|\overrightarrow{AB}|} = \frac{1}{\sqrt{29}} \begin{pmatrix} -5 \\ 2 \end{pmatrix} = \begin{pmatrix} -\frac{5}{\sqrt{29}} \\ \frac{2}{\sqrt{29}} \end{pmatrix} = \begin{pmatrix} -\frac{5\sqrt{29}}{29} \\ \frac{2\sqrt{29}}{29} \end{pmatrix}.$$

2)

$$\overrightarrow{PQ} = \overrightarrow{OQ} - \overrightarrow{OP} = (7\mathbf{i} + 8\mathbf{j}) - (4\mathbf{i} - \mathbf{j}) = 3\mathbf{i} + 9\mathbf{j}.$$
$$|\overrightarrow{PQ}| = \sqrt{3^2 + 9^2} = \sqrt{90} = 3\sqrt{10}.$$

Let $\hat{\mathbf{a}}$ be the unit vector in the direction of $\overrightarrow{PQ}$.

$$\hat{\mathbf{a}} = \frac{\overrightarrow{PQ}}{|\overrightarrow{PQ}|} = \frac{1}{3\sqrt{10}}(3\mathbf{i} + 9\mathbf{j}).$$

Then a vector of length $6\sqrt{10}$ in the direction of $\overrightarrow{PQ}$ is

$$6\sqrt{10} \times \frac{1}{3\sqrt{10}}(3\mathbf{i} + 9\mathbf{j}) = 2(3\mathbf{i} + 9\mathbf{j}) = 6\mathbf{i} + 18\mathbf{j}.$$

3)

$$\overrightarrow{ST} = \overrightarrow{OT} - \overrightarrow{OS} = \begin{pmatrix} \sqrt{96} \\ \sqrt{98} \end{pmatrix} - \begin{pmatrix} \sqrt{150} \\ \sqrt{162} \end{pmatrix} = \begin{pmatrix} 4\sqrt{6} \\ 7\sqrt{2} \end{pmatrix} - \begin{pmatrix} 5\sqrt{6} \\ 9\sqrt{2} \end{pmatrix} = \begin{pmatrix} -\sqrt{6} \\ -2\sqrt{2} \end{pmatrix}.$$
$$|\overrightarrow{ST}| = \sqrt{(-\sqrt{6})^2 + (-2\sqrt{2})^2} = \sqrt{14}.$$

Let $\hat{\mathbf{a}}$ be the unit vector in the direction of $\overrightarrow{ST}$.

$$\hat{\mathbf{a}} = \frac{\overrightarrow{ST}}{|\overrightarrow{ST}|} = \frac{1}{\sqrt{14}} \begin{pmatrix} -\sqrt{6} \\ -2\sqrt{2} \end{pmatrix}.$$

Then a vector of length $\sqrt{126} = 3\sqrt{14}$ in the direction of $\overrightarrow{ST}$ is

$$3\sqrt{14} \times \frac{1}{\sqrt{14}} \begin{pmatrix} -\sqrt{6} \\ -2\sqrt{2} \end{pmatrix} = 3 \begin{pmatrix} -\sqrt{6} \\ -2\sqrt{2} \end{pmatrix} = \begin{pmatrix} -3\sqrt{6} \\ -6\sqrt{2} \end{pmatrix}.$$

4 a)

$$\overrightarrow{AB} = \overrightarrow{OB} - \overrightarrow{OA} = ((2k-3)\mathbf{i} + (3k+1)\mathbf{j}) - ((k-6)\mathbf{i} + (2k+4)\mathbf{j})$$
$$= (k+3)\mathbf{i} + (k-3)\mathbf{j}.$$
$$|\overrightarrow{AB}| = \sqrt{(k+3)^2 + (k-3)^2} = \sqrt{k^2+6k+9+k^2-6k+9} = \sqrt{2k^2+18}.$$

Let $\hat{\mathbf{p}}$ be the unit vector in the direction of $\overrightarrow{AB}$.

$$\hat{\mathbf{p}} = \frac{\overrightarrow{AB}}{|\overrightarrow{AB}|} = \frac{1}{\sqrt{2k^2+18}}((k+3)\mathbf{i} + (k-3)\mathbf{j}).$$

Then a vector of length $\sqrt{4k^2+36}$ in the direction of $\overrightarrow{AB}$ is

$$\mathbf{p} = \sqrt{4k^2+36} \times \frac{1}{\sqrt{2k^2+18}}((k+3)\mathbf{i} + (k-3)\mathbf{j})$$
$$= \frac{\sqrt{4k^2+36}}{\sqrt{2k^2+18}}((k+3)\mathbf{i} + (k-3)\mathbf{j})$$
$$= \sqrt{\frac{4k^2+36}{2k^2+18}}((k+3)\mathbf{i} + (k-3)\mathbf{j})$$
$$= \sqrt{\frac{4(k^2+9)}{2(k^2+9)}}((k+3)\mathbf{i} + (k-3)\mathbf{j})$$
$$= \sqrt{2}((k+3)\mathbf{i} + (k-3)\mathbf{j}).$$

$\therefore \mathbf{p} = (\sqrt{2}k + 3\sqrt{2})\mathbf{i} + (\sqrt{2}k - 3\sqrt{2})\mathbf{j}$.

4 b)

$$|\mathbf{p}| = \sqrt{(\sqrt{2}k + 3\sqrt{2})^2 + (\sqrt{2}k - 3\sqrt{2})^2}$$
$$= \sqrt{(2k^2+12k+18) + (2k^2-12k+18)} = \sqrt{4k^2+36}.$$

If k is a positive integer, then the smallest $|\mathbf{p}|$ can be is when $k = 1$. $\therefore \mathbf{p} = 4\sqrt{2}\mathbf{i} - 2\sqrt{2}\mathbf{j}$.

5 a)

$$\mathbf{a} \cdot \mathbf{b} = \begin{pmatrix} 2 \\ 3 \end{pmatrix} \cdot \begin{pmatrix} 3 \\ 5 \end{pmatrix} = 2 \times 3 + 3 \times 5 = 6 + 15 = 21.$$

5 b)

$$\mathbf{a} \cdot \mathbf{b} = (4\mathbf{i} + \mathbf{j}) \cdot (-2\mathbf{i} + 5\mathbf{j}) = 4 \times (-2) + 1 \times 5 = -8 + 5 = -3.$$

5 c)

$$\mathbf{a} \cdot \mathbf{b} = \begin{pmatrix} 0 \\ -4 \end{pmatrix} \cdot \begin{pmatrix} -3 \\ -6 \end{pmatrix} = 0 \times (-3) + (-4) \times (-6) = 0 + 24 = 24.$$

5 d)

$$\mathbf{a} \cdot \mathbf{b} = (0\mathbf{i} + 9\mathbf{j}) \cdot (6\mathbf{i} - \mathbf{j}) = 0 \times 6 + 9 \times (-1) = 0 - 9 = -9.$$

6)

$$(3p\mathbf{i} - 2\mathbf{j}) \cdot (-5\mathbf{i} - 4p\mathbf{j}) = (3p) \times (-5) + (-2) \times (-4p) = -15p + 8p = -7p.$$

7)

$$\begin{aligned}(8q^3\mathbf{i} + \frac{5}{q}\mathbf{j}) \cdot (-\frac{3}{q^2}\mathbf{i} + 9q^2\mathbf{j}) &= (8q^3) \times (-\frac{3}{q^2}) + (\frac{5}{q}) \times (9q^2) \\ &= -24q + 45q = 21q.\end{aligned}$$

8 a)

$$\cos\theta = \frac{\mathbf{a} \cdot \mathbf{b}}{|\mathbf{a}||\mathbf{b}|} = \frac{1 \times 4 + 9 \times 1}{\sqrt{1^2 + 9^2}\sqrt{4^2 + 1^2}} = \frac{13}{\sqrt{82}\sqrt{17}} = \frac{13\sqrt{1394}}{1394}.$$

$$\theta = \arccos\left(\frac{13\sqrt{1394}}{1394}\right) = 69.6^\circ \text{ (1 DP)}.$$

8 b)

$$\cos\theta = \frac{\mathbf{a} \cdot \mathbf{b}}{|\mathbf{a}||\mathbf{b}|} = \frac{(-2) \times 3 + 10 \times 9}{\sqrt{(-2)^2 + 10^2}\sqrt{3^2 + 9^2}} = \frac{84}{\sqrt{104}\sqrt{90}} = \frac{7\sqrt{65}}{65}.$$

$$\theta = \arccos\left(\frac{7\sqrt{65}}{65}\right) = 29.7^\circ \text{ (1 DP)}.$$

8 c)

$$\cos\theta = \frac{\mathbf{a} \cdot \mathbf{b}}{|\mathbf{a}||\mathbf{b}|} = \frac{(-2) \times 10 + (-11) \times 3}{\sqrt{(-2)^2 + (-11)^2}\sqrt{10^2 + 3^2}} = \frac{-53}{\sqrt{125}\sqrt{109}} = -\frac{53\sqrt{545}}{2725}.$$

$$\theta = \arccos\left(-\frac{53\sqrt{545}}{2725}\right) = 117.0^\circ \text{ (1 DP)}.$$

8 d)

$$\cos\theta = \frac{\mathbf{a} \cdot \mathbf{b}}{|\mathbf{a}||\mathbf{b}|} = \frac{(-8) \times (-10) + 1 \times 2}{\sqrt{(-8)^2 + 1^2}\sqrt{(-10)^2 + 2^2}} = \frac{82}{\sqrt{65}\sqrt{104}} = \frac{41\sqrt{10}}{130}.$$

$$\theta = \arccos\left(\frac{41\sqrt{10}}{130}\right) = 4.2^\circ \text{ (1 DP)}.$$

9)

$$\overrightarrow{BA} = \overrightarrow{OA} - \overrightarrow{OB} = \begin{pmatrix} 2 \\ 4 \end{pmatrix} - \begin{pmatrix} -2 \\ 1 \end{pmatrix} = \begin{pmatrix} 4 \\ 3 \end{pmatrix}.$$

$$\overrightarrow{BC} = \overrightarrow{OC} - \overrightarrow{OB} = \begin{pmatrix} -1 \\ 5 \end{pmatrix} - \begin{pmatrix} -2 \\ 1 \end{pmatrix} = \begin{pmatrix} 1 \\ 4 \end{pmatrix}.$$

$$\cos\theta = \frac{\overrightarrow{BA} \cdot \overrightarrow{BC}}{|\overrightarrow{BA}||\overrightarrow{BC}|} = \frac{4 \times 1 + 3 \times 4}{\sqrt{4^2 + 3^2}\sqrt{1^2 + 4^2}} = \frac{16}{\sqrt{25}\sqrt{17}} = \frac{16\sqrt{17}}{85}.$$

$$\theta = \arccos\left(\frac{16\sqrt{17}}{85}\right) = 39.1^\circ \text{ (1 DP)}.$$

10)

If $\mathbf{a}$ and $\mathbf{b}$ are perpendicular, then $\mathbf{a} \cdot \mathbf{b} = 0$.

11 a)

$$\mathbf{a} \cdot \mathbf{b} = 3 \times (-1) + 1 \times (-3) = -3 - 3 = -6.$$

$\therefore \mathbf{a}$ and $\mathbf{b}$ are not perpendicular.

11 b)

$$\mathbf{a} \cdot \mathbf{b} = 2 \times (-2) + (-5) \times 5 = -4 - 25 = -29.$$

$\therefore \mathbf{a}$ and $\mathbf{b}$ are not perpendicular.

11 c)

$$\mathbf{a} \cdot \mathbf{b} = (-8) \times 2 + 2 \times 8 = -16 + 16 = 0.$$

$\therefore \mathbf{a}$ and $\mathbf{b}$ are perpendicular.

11 d)

$$\mathbf{a} \cdot \mathbf{b} = \sqrt{8} \times \sqrt{72} + (-\sqrt{32}) \times \sqrt{18} = 2\sqrt{2} \times 6\sqrt{2} - 4\sqrt{2} \times 3\sqrt{2} = 24 - 24 = 0.$$

$\therefore \mathbf{a}$ and $\mathbf{b}$ are perpendicular.

12 a)

$$\mathbf{a} \cdot \mathbf{b} = 0 \Rightarrow (2p) \times 5 + 4 \times (-3) = 0 \Rightarrow 10p - 12 = 0 \Rightarrow p = \frac{6}{5}.$$

12 b)

$$\mathbf{a} \cdot \mathbf{b} = 0 \Rightarrow 6 \times (8p) + (-4p) \times \frac{1}{2} = 0 \Rightarrow 48p - 2p = 0 \Rightarrow 46p = 0 \Rightarrow p = 0.$$

12 c)

$$\mathbf{a} \cdot \mathbf{b} = 0 \Rightarrow (4+p) \times (2p) + (3p) \times (1-p) = 0$$
$$\Rightarrow 8p + 2p^2 + 3p - 3p^2 = 0 \Rightarrow 11p - p^2 = 0 \Rightarrow (11-p)p = 0 \Rightarrow p = 0 \text{ or } p = 11.$$

12 d)

$$\mathbf{a} \cdot \mathbf{b} = 0 \Rightarrow (2p-1)(p-2) + (3p-1)(2p+3) = 0$$
$$\Rightarrow 2p^2 - 5p + 2 + 6p^2 + 7p - 3 = 0 \Rightarrow 8p^2 + 2p - 1 = 0$$
$$\Rightarrow (4p-1)(2p+1) = 0 \Rightarrow p = \frac{1}{4} \text{ or } p = -\frac{1}{2}.$$

13)

$$\mathbf{a} \cdot \mathbf{b} = (3k+2)(-6k-6) + (2k+2)(9k+6)$$
$$= -18k^2 - 30k - 12 + 18k^2 + 30k + 12 = 0.$$

$\therefore$ **a** and **b** are perpendicular for all values of k.

14)

$$\mathbf{a} \cdot \mathbf{b} = (k+1)(2-k) + (6k)(4k+1)$$
$$= 2k - k^2 + 2 - k + 24k^2 + 6k = 23k^2 + 7k + 2.$$

If **a** and **b** are perpendicular, then $\mathbf{a} \cdot \mathbf{b} = 0$. Consider $23k^2 + 7k + 2 = 0$.

$$b^2 - 4ac = 7^2 - 4 \times 23 \times 2 = -135 < 0.$$

As $23k^2 + 7k + 2 = 0$ has no real roots, **a** and **b** can never be perpendicular.

15 a)

$$\mathbf{r} = \begin{pmatrix} 3 \\ 1 \end{pmatrix} + \lambda \begin{pmatrix} 2 \\ 1 \end{pmatrix}.$$

15 b)

$$\mathbf{r} = \begin{pmatrix} -2 \\ -5 \end{pmatrix} + \lambda \begin{pmatrix} -1 \\ 9 \end{pmatrix}.$$

15 c)

$$\mathbf{r} = \begin{pmatrix} 0 \\ -8 \end{pmatrix} + \lambda \begin{pmatrix} 0 \\ 1 \end{pmatrix}.$$

15 d)

$$\mathbf{r} = \begin{pmatrix} a \\ b \end{pmatrix} + \lambda \begin{pmatrix} p \\ q \end{pmatrix}.$$

16 a)

$$\begin{pmatrix} 1 \\ 4 \end{pmatrix} - \begin{pmatrix} 2 \\ 9 \end{pmatrix} = \begin{pmatrix} -1 \\ -5 \end{pmatrix}.$$

$$\mathbf{r} = \begin{pmatrix} 1 \\ 4 \end{pmatrix} + \lambda \begin{pmatrix} -1 \\ -5 \end{pmatrix} \text{ or } \mathbf{r} = \begin{pmatrix} 2 \\ 9 \end{pmatrix} + \lambda \begin{pmatrix} -1 \\ -5 \end{pmatrix}$$

$$\text{or } \mathbf{r} = \begin{pmatrix} 1 \\ 4 \end{pmatrix} + \lambda \begin{pmatrix} 1 \\ 5 \end{pmatrix} \text{ or } \mathbf{r} = \begin{pmatrix} 2 \\ 9 \end{pmatrix} + \lambda \begin{pmatrix} 1 \\ 5 \end{pmatrix}.$$

16 b)

$$\begin{pmatrix} 3 \\ 5 \end{pmatrix} - \begin{pmatrix} 5 \\ 15 \end{pmatrix} = \begin{pmatrix} -2 \\ -10 \end{pmatrix}.$$

$$\mathbf{r} = \begin{pmatrix} 3 \\ 5 \end{pmatrix} + \lambda \begin{pmatrix} -2 \\ -10 \end{pmatrix} \text{ or } \mathbf{r} = \begin{pmatrix} 5 \\ 15 \end{pmatrix} + \lambda \begin{pmatrix} -2 \\ -10 \end{pmatrix}$$

$$\text{or } \mathbf{r} = \begin{pmatrix} 3 \\ 5 \end{pmatrix} + \lambda \begin{pmatrix} 1 \\ 5 \end{pmatrix} \text{ or } \mathbf{r} = \begin{pmatrix} 5 \\ 15 \end{pmatrix} + \lambda \begin{pmatrix} 1 \\ 5 \end{pmatrix}.$$

16 c)

$$\begin{pmatrix} -2 \\ 1 \end{pmatrix} - \begin{pmatrix} 0 \\ -5 \end{pmatrix} = \begin{pmatrix} -2 \\ 6 \end{pmatrix}.$$

$$\mathbf{r} = \begin{pmatrix} -2 \\ 1 \end{pmatrix} + \lambda \begin{pmatrix} -2 \\ 6 \end{pmatrix} \text{ or } \mathbf{r} = \begin{pmatrix} 0 \\ -5 \end{pmatrix} + \lambda \begin{pmatrix} -2 \\ 6 \end{pmatrix}$$

$$\text{or } \mathbf{r} = \begin{pmatrix} -2 \\ 1 \end{pmatrix} + \lambda \begin{pmatrix} -1 \\ 3 \end{pmatrix} \text{ or } \mathbf{r} = \begin{pmatrix} 0 \\ -5 \end{pmatrix} + \lambda \begin{pmatrix} -1 \\ 3 \end{pmatrix}.$$

16 d)

$$\begin{pmatrix} -4 \\ 3 \end{pmatrix} - \begin{pmatrix} 4 \\ -13 \end{pmatrix} = \begin{pmatrix} -8 \\ 16 \end{pmatrix}.$$

$$\mathbf{r} = \begin{pmatrix} -4 \\ 3 \end{pmatrix} + \lambda \begin{pmatrix} -8 \\ 16 \end{pmatrix} \text{ or } \mathbf{r} = \begin{pmatrix} 4 \\ -13 \end{pmatrix} + \lambda \begin{pmatrix} -8 \\ 16 \end{pmatrix}$$

$$\text{or } \mathbf{r} = \begin{pmatrix} -4 \\ 3 \end{pmatrix} + \lambda \begin{pmatrix} -1 \\ 2 \end{pmatrix} \text{ or } \mathbf{r} = \begin{pmatrix} 4 \\ -13 \end{pmatrix} + \lambda \begin{pmatrix} -1 \\ 2 \end{pmatrix}.$$

16 e)

$$\begin{pmatrix}2\\9\end{pmatrix}-\begin{pmatrix}-3\\-20\end{pmatrix}=\begin{pmatrix}5\\29\end{pmatrix}.$$

$$\mathbf{r}=\begin{pmatrix}2\\9\end{pmatrix}+\lambda\begin{pmatrix}5\\29\end{pmatrix}\text{ or }\mathbf{r}=\begin{pmatrix}-3\\-20\end{pmatrix}+\lambda\begin{pmatrix}5\\29\end{pmatrix}.$$

16 f)

$$\begin{pmatrix}-4\\-1\end{pmatrix}-\begin{pmatrix}-10\\-7\end{pmatrix}=\begin{pmatrix}6\\6\end{pmatrix}.$$

$$\mathbf{r}=\begin{pmatrix}-4\\-1\end{pmatrix}+\lambda\begin{pmatrix}6\\6\end{pmatrix}\text{ or }\mathbf{r}=\begin{pmatrix}-10\\-7\end{pmatrix}+\lambda\begin{pmatrix}6\\6\end{pmatrix}$$

$$\text{or }\mathbf{r}=\begin{pmatrix}-4\\-1\end{pmatrix}+\lambda\begin{pmatrix}1\\1\end{pmatrix}\text{ or }\mathbf{r}=\begin{pmatrix}-10\\-7\end{pmatrix}+\lambda\begin{pmatrix}1\\1\end{pmatrix}.$$

16 g)

$$\begin{pmatrix}k-2\\2k\end{pmatrix}-\begin{pmatrix}k\\k+1\end{pmatrix}=\begin{pmatrix}-2\\k-1\end{pmatrix}.$$

$$\mathbf{r}=\begin{pmatrix}k\\k+1\end{pmatrix}+\lambda\begin{pmatrix}-2\\k-1\end{pmatrix}\text{ or }\mathbf{r}=\begin{pmatrix}k-2\\2k\end{pmatrix}+\lambda\begin{pmatrix}-2\\k-1\end{pmatrix}$$

$$\text{or }\mathbf{r}=\begin{pmatrix}k\\k+1\end{pmatrix}+\lambda\begin{pmatrix}2\\1-k\end{pmatrix}\text{ or }\mathbf{r}=\begin{pmatrix}k-2\\2k\end{pmatrix}+\lambda\begin{pmatrix}2\\1-k\end{pmatrix}.$$

17)

$$3=5-\lambda\Rightarrow\lambda=2,\qquad 2=-2+2\lambda\Rightarrow\lambda=2.$$

Consistent values of λ. $\therefore (3,2)$ does lie on the line.

18)

$$3=-4+2\lambda\Rightarrow\lambda=\frac{7}{2},\qquad -2=2+\lambda\Rightarrow\lambda=-4.$$

Inconsistent values of λ. $\therefore (3,-2)$ does not lie on the line.

19)

$$-15=-7-2\lambda\Rightarrow\lambda=4,\qquad 15=3+3\lambda\Rightarrow\lambda=4.$$

Consistent values of λ. $\therefore (-15,15)$ lies on the line.

20)

$$4 = -3 + k\lambda \quad (1), \qquad -2 = 9 - 2\lambda \Rightarrow \lambda = \frac{11}{2} \quad (2).$$

Substitute (2) into equation (1):

$$4 = -3 + k \times \frac{11}{2} \Rightarrow k = \frac{14}{11}.$$

21)

$$-3 = 4 + 3\lambda \Rightarrow \lambda = -\frac{7}{3} \quad (1), \qquad -8 = -1 + k\lambda \quad (2).$$

Substitute (1) into equation (2):

$$-8 = -1 + k \times \left(-\frac{7}{3}\right) \Rightarrow k = 3.$$

22 a)

$$4 = k(-5) \Rightarrow k = -\frac{4}{5}, \qquad 1 = k(2) \Rightarrow k = \frac{1}{2}.$$

Inconsistent values of k, so the lines are not parallel (or coincident).

$$-2 + 4\lambda = 1 - 5\mu \Rightarrow 4\lambda + 5\mu = 3 \quad (1)$$

$$-3 + \lambda = 1 + 2\mu \Rightarrow \lambda - 2\mu = 4 \quad (2).$$

Solve (1) and (2): $\lambda = 2$ and $\mu = -1$. Substitute $\lambda = 2$ into l_1: The lines intersect at $(6, -1)$.

22 b)

$$3 = k(-3) \Rightarrow k = -1, \qquad -5 = k(5) \Rightarrow k = -1.$$

Consistent values of k, so the lines are parallel (or coincident). Check if $(4, 6)$ lies on l_2:

$$4 = -5 - 3\mu \Rightarrow \mu = -3, \qquad 6 = 21 + 5\mu \Rightarrow \mu = -3.$$

Consistent values of μ, so $(4, 6)$ lies on l_2. As l_1 and l_2 have the same direction vector, and contain the same point, they must be coincident.

22 c)

$$-2 = k(4) \Rightarrow k = -\frac{1}{2}, \qquad 9 = k(-18) \Rightarrow k = -\frac{1}{2}.$$

Consistent values of k, so the lines are parallel (or coincident). Check if $(6, -1)$ lies on l_2:

$$6 = 1 + 4\mu \Rightarrow \mu = \frac{5}{4}, \qquad -1 = 25 - 18\mu \Rightarrow \mu = \frac{13}{9}.$$

Inconsistent values of μ, so $(6, -1)$ does not lie on l_2. As l_1 and l_2 have the same direction vector, but do not share the same point, they must be parallel and not coincident.

22 d)

$$5 = k(5) \Rightarrow k = 1, \qquad -6 = k(6) \Rightarrow k = -1.$$

Inconsistent values of k, so the lines are not parallel (or coincident).

$$3 + 5\lambda = 4 + 5\mu \Rightarrow 5\lambda - 5\mu = 1 \quad (1)$$

$$-4 - 6\lambda = -3 + 6\mu \Rightarrow -6\lambda - 6\mu = 1 \quad (2).$$

Solve (1) and (2): $\lambda = \frac{1}{60}$ and $\mu = -\frac{11}{60}$. Substitute $\lambda = \frac{1}{60}$ into l_1: The lines intersect at $\left(\frac{37}{12}, -\frac{41}{10}\right)$.

23)

$$\begin{pmatrix} -12 \\ 3 \end{pmatrix} \cdot \begin{pmatrix} 5 \\ 20 \end{pmatrix} = (-12) \times 5 + 3 \times 20 = -60 + 60 = 0.$$

$\therefore l_1$ and l_2 are perpendicular.

24 a)

$$\begin{pmatrix} 5 \\ 3 \end{pmatrix} \cdot \begin{pmatrix} -5 \\ 3 \end{pmatrix} = 5 \times (-5) + 3 \times 3 = -25 + 9 = -16 \neq 0.$$

$\therefore l_1$ and l_2 are not perpendicular.

24 b)

$$\begin{pmatrix} 2 \\ -7 \end{pmatrix} \cdot \begin{pmatrix} -2 \\ 7 \end{pmatrix} = 2 \times (-2) + (-7) \times 7 = -4 - 49 = -53 \neq 0.$$

$\therefore l_1$ and l_2 are not perpendicular.

24 c)

$$\begin{pmatrix} 9 \\ 2 \end{pmatrix} \cdot \begin{pmatrix} 2 \\ 9 \end{pmatrix} = 9 \times 2 + 2 \times 9 = 18 + 18 = 36 \neq 0.$$

$\therefore l_1$ and l_2 are not perpendicular.

24 d)

$$\begin{pmatrix} 3 \\ -11 \end{pmatrix} \cdot \begin{pmatrix} 11 \\ 3 \end{pmatrix} = 3 \times 11 + (-11) \times 3 = 33 - 33 = 0.$$

$\therefore l_1$ and l_2 are perpendicular.

25 a)

$$\begin{pmatrix} k \\ -3 \end{pmatrix} \cdot \begin{pmatrix} 4 \\ 9 \end{pmatrix} = 0 \Rightarrow k \times 4 + (-3) \times 9 = 0$$

$$\Rightarrow 4k - 27 = 0 \Rightarrow k = \frac{27}{4}.$$

25 b)

$$1 + \frac{27}{4}\lambda = 2 + 4\mu \Rightarrow \frac{27}{4}\lambda - 4\mu = 1 \quad (1)$$
$$-100 - 3\lambda = 557 + 9\mu \Rightarrow -3\lambda - 9\mu = 657 \quad (2).$$

Solve (1) and (2): $\lambda = -36$ and $\mu = -61$. Substitute $\lambda = -36$ into l_1: The lines intersect at $(-242, 8)$. So the position vector of the point of intersection is $\begin{pmatrix} -242 \\ 8 \end{pmatrix}$.

26 a)

$$\begin{pmatrix} -5 \\ k \end{pmatrix} \cdot \begin{pmatrix} 20 \\ 20 \end{pmatrix} = 0 \Rightarrow (-5) \times 20 + k \times 20 = 0$$
$$\Rightarrow -100 + 20k = 0 \Rightarrow k = 5.$$

26 b)

$$4 - 5\lambda = -2 - \mu \Rightarrow -5\lambda + \mu = -6 \quad (1)$$
$$8 + 4\lambda = 14 + 20\mu \Rightarrow 5\lambda - 20\mu = 6 \quad (2).$$

Solve (1) and (2): $\lambda = \frac{6}{5}$ and $\mu = 0$. Substitute $\lambda = \frac{6}{5}$ into l_1: The lines intersect at $P(-2, 14)$.

26 c)

$$\overrightarrow{OP} = \begin{pmatrix} -2 \\ 14 \end{pmatrix}, \qquad |\overrightarrow{OP}| = \sqrt{(-2)^2 + 14^2} = \sqrt{200} = 10\sqrt{2}.$$

Let $\hat{\mathbf{p}}$ be a unit vector in the direction of $\overrightarrow{OP}$.

$$\hat{\mathbf{p}} = \frac{\overrightarrow{OP}}{|\overrightarrow{OP}|} = \frac{1}{10\sqrt{2}} \begin{pmatrix} -2 \\ 14 \end{pmatrix} = \frac{\sqrt{2}}{20} \begin{pmatrix} -2 \\ 14 \end{pmatrix} = \begin{pmatrix} -\frac{\sqrt{2}}{10} \\ \frac{7\sqrt{2}}{10} \end{pmatrix}.$$

27)

If l_1 and l_2 intersect and are not coincident, then they cannot be parallel. If they were parallel then $\begin{pmatrix} 15 \\ k \end{pmatrix} = p \begin{pmatrix} 3 \\ 2 \end{pmatrix}$ for some constant p.

$$15 = 3p \Rightarrow p = 5, \qquad k = 5 \times 2 = 10.$$

So k can be any real value except 10, i.e. $k \neq 10$.

28 a)

$$5 + 3\lambda = 4 + 10\mu \Rightarrow 3\lambda - 10\mu = -1 \quad (1)$$
$$420 - 10\lambda = 120 - 3\mu \Rightarrow -10\lambda + 3\mu = -300 \quad (2).$$

Solve (1) and (2): $\lambda = 33$ and $\mu = 10$. Substitute $\lambda = 33$ into l_1: The lines intersect at $(104, 90)$.

28 b)

$$\begin{pmatrix}104\\90\end{pmatrix}-\begin{pmatrix}90\\100\end{pmatrix}=\begin{pmatrix}14\\-10\end{pmatrix}.$$

$\therefore l_3 : \mathbf{r}=\begin{pmatrix}90\\100\end{pmatrix}+\gamma\begin{pmatrix}14\\-10\end{pmatrix}$ or $l_3 : \mathbf{r}=\begin{pmatrix}90\\100\end{pmatrix}+\gamma\begin{pmatrix}7\\-5\end{pmatrix}$.

29 a)

$$\cos\theta=\frac{\mathbf{d}_1\cdot\mathbf{d}_2}{|\mathbf{d}_1||\mathbf{d}_2|}=\frac{\begin{pmatrix}2\\5\end{pmatrix}\cdot\begin{pmatrix}3\\9\end{pmatrix}}{\left|\begin{pmatrix}2\\5\end{pmatrix}\right|\left|\begin{pmatrix}3\\9\end{pmatrix}\right|}=\frac{2\times3+5\times9}{\sqrt{2^2+5^2}\sqrt{3^2+9^2}}$$

$$=\frac{51}{\sqrt{29}\sqrt{90}}=\frac{17\sqrt{290}}{290},\qquad\theta=\arccos\left(\frac{17\sqrt{290}}{290}\right)=3.37^\circ\text{ (1 DP)}.$$

29 b)

$$\cos\theta=\frac{\begin{pmatrix}3\\4\end{pmatrix}\cdot\begin{pmatrix}-5\\11\end{pmatrix}}{\left|\begin{pmatrix}3\\4\end{pmatrix}\right|\left|\begin{pmatrix}-5\\11\end{pmatrix}\right|}=\frac{3\times(-5)+4\times11}{\sqrt{3^2+4^2}\sqrt{(-5)^2+11^2}}$$

$$=\frac{29}{5\sqrt{146}}=\frac{29\sqrt{146}}{730},\qquad\theta=\arccos\left(\frac{29\sqrt{146}}{730}\right)=61.3^\circ\text{ (1 DP)}.$$

29 c)

$$\cos\theta=\frac{\begin{pmatrix}-2\\3\end{pmatrix}\cdot\begin{pmatrix}5\\1\end{pmatrix}}{\left|\begin{pmatrix}-2\\3\end{pmatrix}\right|\left|\begin{pmatrix}5\\1\end{pmatrix}\right|}=\frac{(-2)\times5+3\times1}{\sqrt{(-2)^2+3^2}\sqrt{5^2+1^2}}=-\frac{7}{\sqrt{13}\sqrt{26}}=-\frac{7\sqrt{2}}{26}.$$

$$\theta=\arccos\left(-\frac{7\sqrt{2}}{26}\right)=112.380\ldots^\circ.$$

Acute angle $=180^\circ-112.380\ldots^\circ=67.6^\circ$ (1 DP).

29 d)

$$\cos\theta=\frac{\begin{pmatrix}-6\\13\end{pmatrix}\cdot\begin{pmatrix}2\\-3\end{pmatrix}}{\left|\begin{pmatrix}-6\\13\end{pmatrix}\right|\left|\begin{pmatrix}2\\-3\end{pmatrix}\right|}=\frac{(-6)\times2+13\times(-3)}{\sqrt{(-6)^2+13^2}\sqrt{2^2+(-3)^2}}=-\frac{51}{\sqrt{205}\sqrt{13}}=-\frac{51\sqrt{2665}}{2665}.$$

$$\theta=\arccos\left(-\frac{51\sqrt{2665}}{2665}\right)=171.085\ldots^\circ.$$

Acute angle $=180^\circ-171.085\ldots^\circ=8.9^\circ$ (1 DP).

30 a)

$$\cos\theta = \frac{\mathbf{d}_1 \cdot \mathbf{d}_2}{|\mathbf{d}_1||\mathbf{d}_2|} = \frac{\begin{pmatrix} 3 \\ -5 \end{pmatrix} \cdot \begin{pmatrix} 1 \\ 0 \end{pmatrix}}{\left|\begin{pmatrix} 3 \\ -5 \end{pmatrix}\right| \left|\begin{pmatrix} 1 \\ 0 \end{pmatrix}\right|} = \frac{3}{\sqrt{34}\sqrt{1}} = \frac{3\sqrt{34}}{34}.$$

$$\theta = \arccos\left(\frac{3\sqrt{34}}{34}\right) = 59.0^\circ \text{ (1 DP)}.$$

30 b)

$$\cos\theta = \frac{\mathbf{d}_1 \cdot \mathbf{d}_2}{|\mathbf{d}_1||\mathbf{d}_2|} = \frac{\begin{pmatrix} 3 \\ -5 \end{pmatrix} \cdot \begin{pmatrix} 0 \\ 1 \end{pmatrix}}{\left|\begin{pmatrix} 3 \\ -5 \end{pmatrix}\right| \left|\begin{pmatrix} 0 \\ 1 \end{pmatrix}\right|} = -\frac{5}{\sqrt{34}\sqrt{1}} = -\frac{5\sqrt{34}}{34}.$$

$$\theta = \arccos\left(-\frac{5\sqrt{34}}{34}\right) = 149.036\ldots^\circ.$$

Acute angle $= 180^\circ - 149.036\ldots^\circ = 31.0^\circ$ (1 DP).

31)

$$\cos\theta = \frac{\mathbf{d}_1 \cdot \mathbf{d}_2}{|\mathbf{d}_1||\mathbf{d}_2|} = \frac{\begin{pmatrix} 4 \\ -1 \end{pmatrix} \cdot \begin{pmatrix} 3 \\ 4 \end{pmatrix}}{\left|\begin{pmatrix} 4 \\ -1 \end{pmatrix}\right| \left|\begin{pmatrix} 3 \\ 4 \end{pmatrix}\right|} = \frac{4 \times 3 + (-1) \times 4}{\sqrt{4^2 + (-1)^2}\sqrt{3^2 + 4^2}}$$

$$= \frac{8}{\sqrt{17} \times 5} = \frac{8\sqrt{17}}{85}.$$

$$\theta = \arccos\left(\frac{8\sqrt{17}}{85}\right) = 67.166\ldots^\circ.$$

Obtuse angle $= 180^\circ - 67.166\ldots^\circ = 112.8^\circ$ (1 DP).

32)

$$\begin{pmatrix} 3 & 2 \\ -1 & 5 \end{pmatrix} \begin{pmatrix} 3+\lambda \\ -2+4\lambda \end{pmatrix} = \begin{pmatrix} 3(3+\lambda) + 2(-2+4\lambda) \\ -1(3+\lambda) + 5(-2+4\lambda) \end{pmatrix}$$

$$= \begin{pmatrix} 5+11\lambda \\ -13+19\lambda \end{pmatrix} = \begin{pmatrix} 5 \\ -13 \end{pmatrix} + \lambda \begin{pmatrix} 11 \\ 19 \end{pmatrix}.$$

$\therefore l_2 : \mathbf{r} = \begin{pmatrix} 5 \\ -13 \end{pmatrix} + \lambda \begin{pmatrix} 11 \\ 19 \end{pmatrix}$.

33)

$$\begin{pmatrix} 4 & 1 \\ 2 & -3 \end{pmatrix} \begin{pmatrix} 6+2\lambda \\ -3-7\lambda \end{pmatrix} = \begin{pmatrix} 4(6+2\lambda)+1(-3-7\lambda) \\ 2(6+2\lambda)-3(-3-7\lambda) \end{pmatrix}$$
$$= \begin{pmatrix} 21+\lambda \\ 21+25\lambda \end{pmatrix} = \begin{pmatrix} 21 \\ 21 \end{pmatrix} + \lambda \begin{pmatrix} 1 \\ 25 \end{pmatrix}.$$

$\therefore l_2 : \mathbf{r} = \begin{pmatrix} 21 \\ 21 \end{pmatrix} + \lambda \begin{pmatrix} 1 \\ 25 \end{pmatrix}.$

34 a)

$$\begin{pmatrix} 5 & 0 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} -9+5\lambda \\ 4-8\lambda \end{pmatrix} = \begin{pmatrix} 5(-9+5\lambda)+0(4-8\lambda) \\ 0(-9+5\lambda)+1(4-8\lambda) \end{pmatrix}$$
$$= \begin{pmatrix} -45+25\lambda \\ 4-8\lambda \end{pmatrix} = \begin{pmatrix} -45 \\ 4 \end{pmatrix} + \lambda \begin{pmatrix} 25 \\ -8 \end{pmatrix}.$$

$\therefore l_2 : \mathbf{r} = \begin{pmatrix} -45 \\ 4 \end{pmatrix} + \lambda \begin{pmatrix} 25 \\ -8 \end{pmatrix}.$

34 b)

$$\begin{pmatrix} 2 & 0 \\ 0 & 2 \end{pmatrix} \begin{pmatrix} -9+5\lambda \\ 4-8\lambda \end{pmatrix} = \begin{pmatrix} 2(-9+5\lambda)+0(4-8\lambda) \\ 0(-9+5\lambda)+2(4-8\lambda) \end{pmatrix}$$
$$= \begin{pmatrix} -18+10\lambda \\ 8-16\lambda \end{pmatrix} = \begin{pmatrix} -18 \\ 8 \end{pmatrix} + \lambda \begin{pmatrix} 10 \\ -16 \end{pmatrix}.$$

$\therefore l_2 : \mathbf{r} = \begin{pmatrix} -18 \\ 8 \end{pmatrix} + \lambda \begin{pmatrix} 10 \\ -16 \end{pmatrix}$ or $l_2 : \mathbf{r} = \begin{pmatrix} -18 \\ 8 \end{pmatrix} + \lambda \begin{pmatrix} 5 \\ -8 \end{pmatrix}.$

34 c)

$$\begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \begin{pmatrix} -9+5\lambda \\ 4-8\lambda \end{pmatrix} = \begin{pmatrix} 1(-9+5\lambda)+0(4-8\lambda) \\ 0(-9+5\lambda)-1(4-8\lambda) \end{pmatrix}$$
$$= \begin{pmatrix} -9+5\lambda \\ -4+8\lambda \end{pmatrix} = \begin{pmatrix} -9 \\ -4 \end{pmatrix} + \lambda \begin{pmatrix} 5 \\ 8 \end{pmatrix}.$$

$\therefore l_2 : \mathbf{r} = \begin{pmatrix} -9 \\ -4 \end{pmatrix} + \lambda \begin{pmatrix} 5 \\ 8 \end{pmatrix}.$

34 d)

$$\begin{pmatrix} -1 & 0 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} -9+5\lambda \\ 4-8\lambda \end{pmatrix} = \begin{pmatrix} -1(-9+5\lambda)+0(4-8\lambda) \\ 0(-9+5\lambda)+1(4-8\lambda) \end{pmatrix}$$
$$= \begin{pmatrix} 9-5\lambda \\ 4-8\lambda \end{pmatrix} = \begin{pmatrix} 9 \\ 4 \end{pmatrix} + \lambda \begin{pmatrix} -5 \\ -8 \end{pmatrix}.$$

$\therefore l_2 : \mathbf{r} = \begin{pmatrix} 9 \\ 4 \end{pmatrix} + \lambda \begin{pmatrix} -5 \\ -8 \end{pmatrix}$ or $l_2 : \mathbf{r} = \begin{pmatrix} 9 \\ 4 \end{pmatrix} + \lambda \begin{pmatrix} 5 \\ 8 \end{pmatrix}.$

34 e)

$$\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}\begin{pmatrix} -9+5\lambda \\ 4-8\lambda \end{pmatrix} = \begin{pmatrix} 0(-9+5\lambda)+1(4-8\lambda) \\ 1(-9+5\lambda)+0(4-8\lambda) \end{pmatrix}$$
$$= \begin{pmatrix} 4-8\lambda \\ -9+5\lambda \end{pmatrix} = \begin{pmatrix} 4 \\ -9 \end{pmatrix} + \lambda\begin{pmatrix} -8 \\ 5 \end{pmatrix}.$$

$\therefore l_2 : \mathbf{r} = \begin{pmatrix} 4 \\ -9 \end{pmatrix} + \lambda\begin{pmatrix} -8 \\ 5 \end{pmatrix}$.

34 f)

$$\begin{pmatrix} -1 & 0 \\ 0 & -1 \end{pmatrix}\begin{pmatrix} -9+5\lambda \\ 4-8\lambda \end{pmatrix} = \begin{pmatrix} -1(-9+5\lambda)+0(4-8\lambda) \\ 0(-9+5\lambda)-1(4-8\lambda) \end{pmatrix}$$
$$= \begin{pmatrix} 9-5\lambda \\ -4+8\lambda \end{pmatrix} = \begin{pmatrix} 9 \\ -4 \end{pmatrix} + \lambda\begin{pmatrix} -5 \\ 8 \end{pmatrix}.$$

$\therefore l_2 : \mathbf{r} = \begin{pmatrix} 9 \\ -4 \end{pmatrix} + \lambda\begin{pmatrix} -5 \\ 8 \end{pmatrix}$.

35)

Using a non-Vector method:
The line passing through $(0,0)$ which is perpendicular to $y = 2x+5$ is $y = -\frac{1}{2}x$. $y = 2x+5$ and $y = -\frac{1}{2}x$ intersect at $(-2,1)$. Distance between $(0,0)$ and $(-2,1)$ is $\sqrt{(-2-0)^2+(1-0)^2} = \sqrt{5}$ units.

Alternative Method (using Vectors):
Let l be the line $\mathbf{r} = \begin{pmatrix} 0 \\ 5 \end{pmatrix} + \lambda\begin{pmatrix} 1 \\ 2 \end{pmatrix}$. Let P be a point on the line, so $\overrightarrow{OP} = \begin{pmatrix} \lambda \\ 5+2\lambda \end{pmatrix}$.

$$\overrightarrow{OP}\cdot\begin{pmatrix} 1 \\ 2 \end{pmatrix} = 0 \Rightarrow \begin{pmatrix} \lambda \\ 5+2\lambda \end{pmatrix}\cdot\begin{pmatrix} 1 \\ 2 \end{pmatrix} = 0$$
$$\Rightarrow \lambda\times 1 + (5+2\lambda)\times 2 = 0 \Rightarrow \lambda + 10 + 4\lambda = 0 \Rightarrow 5\lambda + 10 = 0 \Rightarrow \lambda = -2.$$

So $\overrightarrow{OP} = \begin{pmatrix} -2 \\ 5+2(-2) \end{pmatrix} = \begin{pmatrix} -2 \\ 1 \end{pmatrix}$.

$$|\overrightarrow{OP}| = \sqrt{(-2)^2+1^2} = \sqrt{5} \text{ units.}$$

Alternative Method (using the Vector Product):
Convert to a 3D problem. Let $\mathbf{p} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}$. Let A be a known point on the line so that its position vector is $\mathbf{a} = \begin{pmatrix} 0 \\ 5 \\ 0 \end{pmatrix}$. Let $\mathbf{d} = \begin{pmatrix} 1 \\ 2 \\ 0 \end{pmatrix}$. Then

$$h = \frac{|\mathbf{d}\times(\mathbf{p}-\mathbf{a})|}{|\mathbf{d}|} = \frac{\left|\begin{pmatrix} 1 \\ 2 \\ 0 \end{pmatrix}\times\left(\begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}-\begin{pmatrix} 0 \\ 5 \\ 0 \end{pmatrix}\right)\right|}{\left|\begin{pmatrix} 1 \\ 2 \\ 0 \end{pmatrix}\right|} = \frac{\left|\begin{pmatrix} 1 \\ 2 \\ 0 \end{pmatrix}\times\begin{pmatrix} 0 \\ -5 \\ 0 \end{pmatrix}\right|}{\left|\begin{pmatrix} 1 \\ 2 \\ 0 \end{pmatrix}\right|}.$$

$$\begin{pmatrix}1\\2\\0\end{pmatrix}\times\begin{pmatrix}0\\-5\\0\end{pmatrix}=\begin{vmatrix}\mathbf{i}&\mathbf{j}&\mathbf{k}\\1&2&0\\0&-5&0\end{vmatrix}$$

$$=(2\times 0-0\times(-5))\mathbf{i}-(1\times 0-0\times 0)\mathbf{j}+(1\times(-5)-2\times 0)\mathbf{k}=0\mathbf{i}+0\mathbf{j}-5\mathbf{k}.$$

$$\therefore h=\frac{\left|\begin{pmatrix}0\\0\\-5\end{pmatrix}\right|}{\left|\begin{pmatrix}1\\2\\0\end{pmatrix}\right|}=\frac{\sqrt{0^2+0^2+(-5)^2}}{\sqrt{1^2+2^2+0^2}}=\frac{5}{\sqrt{5}}=\sqrt{5}\text{ units.}$$

36)

Using a non-Vector method:
The line passing through $(3,-5)$ which is perpendicular to $y=8x+1$ is

$$y+5=-\frac{1}{8}(x-3)\Rightarrow y=-\frac{1}{8}x-\frac{37}{8}.$$

$y=8x+1$ and $y=-\frac{1}{8}x-\frac{37}{8}$ intersect at $(-\frac{9}{13},-\frac{59}{13})$. Distance between $(3,-5)$ and $(-\frac{9}{13},-\frac{59}{13})$ is

$$\sqrt{(-\frac{9}{13}-3)^2+(-\frac{59}{13}+5)^2}=\sqrt{\frac{180}{13}}=\frac{6\sqrt{65}}{13}\text{ units.}$$

Alternative Method (using Vectors):

Let A be the point with coordinates $(3,-5)$. Let l be the line $\mathbf{r}=\begin{pmatrix}0\\1\end{pmatrix}+\lambda\begin{pmatrix}1\\8\end{pmatrix}$. Let P be a point on the line, so $\overrightarrow{OP}=\begin{pmatrix}\lambda\\1+8\lambda\end{pmatrix}$.

$$\overrightarrow{AP}=\overrightarrow{OP}-\overrightarrow{OA}=\begin{pmatrix}\lambda\\1+8\lambda\end{pmatrix}-\begin{pmatrix}3\\-5\end{pmatrix}=\begin{pmatrix}\lambda-3\\6+8\lambda\end{pmatrix}.$$

$$\overrightarrow{AP}\cdot\begin{pmatrix}1\\8\end{pmatrix}=0\Rightarrow(\lambda-3)\times 1+(6+8\lambda)\times 8=0$$

$$\Rightarrow\lambda-3+48+64\lambda=0\Rightarrow 65\lambda+45=0\Rightarrow\lambda=-\frac{9}{13}.$$

So $\overrightarrow{AP}=\begin{pmatrix}(-\frac{9}{13})-3\\6+8(-\frac{9}{13})\end{pmatrix}=\begin{pmatrix}-\frac{48}{13}\\\frac{6}{13}\end{pmatrix}$.

$$|\overrightarrow{AP}|=\sqrt{(-\frac{48}{13})^2+(\frac{6}{13})^2}=\frac{6\sqrt{65}}{13}\text{ units.}$$

Alternative Method (using the Vector Product):

Convert to a 3D problem. Let $\mathbf{p}=\begin{pmatrix}3\\-5\\0\end{pmatrix}$. Let A be a known point on the line so that its position

vector is $\mathbf{a} = \begin{pmatrix}0\\1\\0\end{pmatrix}$. Let $\mathbf{d} = \begin{pmatrix}1\\8\\0\end{pmatrix}$. Then

$$h = \frac{|\mathbf{d} \times (\mathbf{p} - \mathbf{a})|}{|\mathbf{d}|} = \frac{\left|\begin{pmatrix}1\\8\\0\end{pmatrix} \times \begin{pmatrix}3\\-6\\0\end{pmatrix}\right|}{\left|\begin{pmatrix}1\\8\\0\end{pmatrix}\right|}.$$

$$\begin{pmatrix}1\\8\\0\end{pmatrix} \times \begin{pmatrix}3\\-6\\0\end{pmatrix} = \begin{vmatrix}\mathbf{i} & \mathbf{j} & \mathbf{k}\\1 & 8 & 0\\3 & -6 & 0\end{vmatrix} = 0\mathbf{i} + 0\mathbf{j} - 30\mathbf{k}.$$

$$\therefore h = \frac{\left|\begin{pmatrix}0\\0\\-30\end{pmatrix}\right|}{\left|\begin{pmatrix}1\\8\\0\end{pmatrix}\right|} = \frac{\sqrt{0^2+0^2+(-30)^2}}{\sqrt{1^2+8^2+0^2}} = \frac{30}{\sqrt{65}} = \frac{6\sqrt{65}}{13} \text{ units.}$$

37)

Using a non-Vector method:
The line has cartesian equation $y - 0 = \frac{1}{2}(x+1) = y = \frac{1}{2}x + \frac{1}{2}$. The line passing through $(4, -2)$ which is perpendicular to $y = \frac{1}{2}x + \frac{1}{2}$ is

$$y + 2 = -2(x - 4) \Rightarrow y = -2x + 6.$$

$y = \frac{1}{2}x + \frac{1}{2}$ and $y = -2x + 6$ intersect at $(\frac{11}{5}, \frac{8}{5})$. Distance between $(\frac{11}{5}, \frac{8}{5})$ and $(4, -2)$ is

$$\sqrt{(4 - \frac{11}{5})^2 + (-2 - \frac{8}{5})^2} = \sqrt{\frac{81}{5}} = \frac{9\sqrt{5}}{5} \text{ units.}$$

Alternative Method (using Vectors):

Let A be the point with coordinates $(4, -2)$. Let P be a point on the line, so $\overrightarrow{OP} = \begin{pmatrix}-1+2\lambda\\\lambda\end{pmatrix}$.

$$\overrightarrow{AP} = \overrightarrow{OP} - \overrightarrow{OA} = \begin{pmatrix}-1+2\lambda\\\lambda\end{pmatrix} - \begin{pmatrix}4\\-2\end{pmatrix} = \begin{pmatrix}-5+2\lambda\\\lambda+2\end{pmatrix}.$$

$$\overrightarrow{AP} \cdot \begin{pmatrix}2\\1\end{pmatrix} = 0 \Rightarrow (-5+2\lambda) \times 2 + (\lambda+2) \times 1 = 0$$

$$\Rightarrow -10 + 4\lambda + \lambda + 2 = 0 \Rightarrow 5\lambda - 8 = 0 \Rightarrow \lambda = \frac{8}{5}.$$

So $\overrightarrow{AP} = \begin{pmatrix}-5+2(\frac{8}{5})\\\frac{8}{5}+2\end{pmatrix} = \begin{pmatrix}-\frac{9}{5}\\\frac{18}{5}\end{pmatrix}$.

$$|\overrightarrow{AP}| = \sqrt{(-\frac{9}{5})^2 + (\frac{18}{5})^2} = \frac{9\sqrt{5}}{5} \text{ units.}$$

Alternative Method (using the Vector Product):

Convert to a 3D problem. Let $\mathbf{p} = \begin{pmatrix} 4 \\ -2 \\ 0 \end{pmatrix}$. Let A be a known point on the line so that its position vector is $\mathbf{a} = \begin{pmatrix} -1 \\ 0 \\ 0 \end{pmatrix}$. Let $\mathbf{d} = \begin{pmatrix} 2 \\ 1 \\ 0 \end{pmatrix}$. Then

$$h = \frac{|\mathbf{d} \times (\mathbf{p} - \mathbf{a})|}{|\mathbf{d}|} = \frac{\left| \begin{pmatrix} 2 \\ 1 \\ 0 \end{pmatrix} \times \begin{pmatrix} 5 \\ -2 \\ 0 \end{pmatrix} \right|}{\left| \begin{pmatrix} 2 \\ 1 \\ 0 \end{pmatrix} \right|}.$$

$$\begin{pmatrix} 2 \\ 1 \\ 0 \end{pmatrix} \times \begin{pmatrix} 5 \\ -2 \\ 0 \end{pmatrix} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 2 & 1 & 0 \\ 5 & -2 & 0 \end{vmatrix} = 0\mathbf{i} + 0\mathbf{j} - 9\mathbf{k}.$$

$$\therefore h = \frac{\left| \begin{pmatrix} 0 \\ 0 \\ -9 \end{pmatrix} \right|}{\left| \begin{pmatrix} 2 \\ 1 \\ 0 \end{pmatrix} \right|} = \frac{9}{\sqrt{5}} = \frac{9\sqrt{5}}{5} \text{ units.}$$

38)

Using a non-Vector method:
The line has cartesian equation $y + 2 = -\frac{3}{5}(x - 4) = y = -\frac{3}{5}x + \frac{2}{5}$. The line passing through $(1, k)$ which is perpendicular to $y = -\frac{3}{5}x + \frac{2}{5}$ is

$$y - k = \frac{5}{3}(x - 1) \Rightarrow y = \frac{5}{3}x + k - \frac{5}{3}.$$

These lines intersect when

$$-\frac{3}{5}x + \frac{2}{5} = \frac{5}{3}x + k - \frac{5}{3} \Rightarrow \frac{34}{15}x = \frac{31}{15} - k \Rightarrow x = \frac{31}{34} - \frac{15}{34}k.$$

$$\therefore y = -\frac{3}{5}(\frac{31}{34} - \frac{15}{34}k) + \frac{2}{5} = \frac{9}{34}k - \frac{5}{34}.$$

The lines intersect at $\left(\frac{31}{34} - \frac{15}{34}k, \frac{9}{34}k - \frac{5}{34}\right)$. Distance between this point and $(1, k)$ is 6664:

$$\sqrt{(1 - (\frac{31}{34} - \frac{15}{34}k))^2 + (k - (\frac{9}{34}k - \frac{5}{34}))^2} = 14\sqrt{34}$$

$$\Rightarrow (\frac{15}{34}k + \frac{3}{34})^2 + (\frac{25k}{34} + \frac{5}{34})^2 = 14\sqrt{34}$$

$$\Rightarrow (\frac{1}{34})^2(15k + 3)^2 + (\frac{1}{34})^2(25k + 5)^2 = 6664$$

$$\Rightarrow (15k + 3)^2 + (25k + 5)^2 = 7703584$$

$$\Rightarrow 225k^2 + 90k + 9 + 625k^2 + 250k + 25 = 7703584$$

$$\Rightarrow 850k^2 + 340k - 7703550 = 0 \Rightarrow 5k^2 + 2k - 45315 = 0$$
$$\Rightarrow (5k + 477)(k - 95) = 0 \Rightarrow k = -\frac{477}{5} \text{ or } k = 95.$$

As k is an integer, $k = 95$.

Alternative Method (using Vectors):

Let A be the point with coordinates $(1, k)$. Let P be a point on the line, so $\overrightarrow{OP} = \begin{pmatrix} 4 - 5\lambda \\ -2 + 3\lambda \end{pmatrix}$.

$$\overrightarrow{AP} = \overrightarrow{OP} - \overrightarrow{OA} = \begin{pmatrix} 4 - 5\lambda \\ -2 + 3\lambda \end{pmatrix} - \begin{pmatrix} 1 \\ k \end{pmatrix} = \begin{pmatrix} 3 - 5\lambda \\ -2 - k + 3\lambda \end{pmatrix}.$$

$$\overrightarrow{AP} \cdot \begin{pmatrix} -5 \\ 3 \end{pmatrix} = 0 \Rightarrow (3 - 5\lambda)(-5) + (-2 - k + 3\lambda)3 = 0$$

$$\Rightarrow -15 + 25\lambda - 6 - 3k + 9\lambda = 0 \Rightarrow 34\lambda - 21 - 3k = 0 \Rightarrow \lambda = \frac{21 + 3k}{34}.$$

So

$$\overrightarrow{AP} = \begin{pmatrix} 3 - 5(\frac{21+3k}{34}) \\ -2 - k + 3(\frac{21+3k}{34}) \end{pmatrix} = \begin{pmatrix} -\frac{15}{34}k - \frac{3}{34} \\ -\frac{25}{34}k - \frac{5}{34} \end{pmatrix}.$$

$$|\overrightarrow{AP}| = 14\sqrt{34} \Rightarrow \sqrt{(-\frac{15}{34}k - \frac{3}{34})^2 + (-\frac{25}{34}k - \frac{5}{34})^2} = 14\sqrt{34}$$
$$\Rightarrow (\frac{1}{34})^2(15k + 3)^2 + (\frac{1}{34})^2(25k + 5)^2 = 6664$$
$$\Rightarrow (15k + 3)^2 + (25k + 5)^2 = 7703584$$
$$\Rightarrow 225k^2 + 90k + 9 + 625k^2 + 250k + 25 = 7703584$$
$$\Rightarrow 850k^2 + 340k - 7703550 = 0 \Rightarrow 5k^2 + 2k - 45315 = 0$$
$$\Rightarrow (5k + 477)(k - 95) = 0 \Rightarrow k = -\frac{477}{5} \text{ or } k = 95.$$

As k is an integer, $k = 95$.

Alternative Method (using the Vector Product):

Convert to a 3D problem. Let $\mathbf{p} = \begin{pmatrix} 1 \\ k \\ 0 \end{pmatrix}$. Let A be a known point on the line so that its position vector is $\mathbf{a} = \begin{pmatrix} 4 \\ -2 \\ 0 \end{pmatrix}$. Let $\mathbf{d} = \begin{pmatrix} -5 \\ 3 \\ 0 \end{pmatrix}$. Then

$$h = \frac{|\mathbf{d} \times (\mathbf{p} - \mathbf{a})|}{|\mathbf{d}|} = \frac{\left| \begin{pmatrix} -5 \\ 3 \\ 0 \end{pmatrix} \times \begin{pmatrix} -3 \\ k + 2 \\ 0 \end{pmatrix} \right|}{\left| \begin{pmatrix} -5 \\ 3 \\ 0 \end{pmatrix} \right|}.$$

$$\begin{pmatrix} -5 \\ 3 \\ 0 \end{pmatrix} \times \begin{pmatrix} -3 \\ k + 2 \\ 0 \end{pmatrix} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ -5 & 3 & 0 \\ -3 & k + 2 & 0 \end{vmatrix} = 0\mathbf{i} + 0\mathbf{j} + (-5k - 1)\mathbf{k}.$$

$$\therefore h = 14\sqrt{34} \Rightarrow \frac{\sqrt{(-5k - 1)^2}}{\sqrt{34}} = 14\sqrt{34} \Rightarrow \sqrt{25k^2 + 10k + 1} = 476$$

$$\Rightarrow 25k^2 + 10k + 1 = 226576 \Rightarrow 25k^2 + 10k - 226575 = 0$$

$$\Rightarrow 5k^2 + 2k - 45315 = 0 \Rightarrow (5k + 477)(k - 95) = 0$$

$$\Rightarrow k = -\frac{477}{5} \text{ or } k = 95.$$

As k is an integer, $k = 95$.

39)

Using a non-Vector method:
Choose a point on l_1: $(0, 2)$. The line passing through $(0, 2)$ which is perpendicular to $y = x + 2$ is $y - 2 = -1(x - 0) \Rightarrow y = -x + 2$. $y = -x + 2$ and $y = x + 8$ intersect at $(-3, 5)$. Distance between $(-3, 5)$ and $(0, 2)$ is

$$\sqrt{(0+3)^2 + (2-5)^2} = \sqrt{18} = 3\sqrt{2} \text{ units.}$$

Alternative Method (using Vectors):

Choose a point on l_1 so let A be the point with coordinates $(0, 2)$. Let l_2 be $\mathbf{r} = \begin{pmatrix} 0 \\ 8 \end{pmatrix} + \lambda \begin{pmatrix} 1 \\ 1 \end{pmatrix}$. Let P be a point on the line, so $\overrightarrow{OP} = \begin{pmatrix} \lambda \\ 8 + \lambda \end{pmatrix}$.

$$\overrightarrow{AP} = \overrightarrow{OP} - \overrightarrow{OA} = \begin{pmatrix} \lambda \\ 8 + \lambda \end{pmatrix} - \begin{pmatrix} 0 \\ 2 \end{pmatrix} = \begin{pmatrix} \lambda \\ 6 + \lambda \end{pmatrix}.$$

$$\overrightarrow{AP} \cdot \begin{pmatrix} 1 \\ 1 \end{pmatrix} = 0 \Rightarrow \lambda \times 1 + (6 + \lambda) \times 1 = 0 \Rightarrow \lambda + 6 + \lambda = 0$$

$$\Rightarrow 2\lambda + 6 = 0 \Rightarrow \lambda = -3.$$

So $\overrightarrow{AP} = \begin{pmatrix} -3 \\ 6 + (-3) \end{pmatrix} = \begin{pmatrix} -3 \\ 3 \end{pmatrix}$ and

$$|\overrightarrow{AP}| = \sqrt{(-3)^2 + 3^2} = \sqrt{18} = 3\sqrt{2} \text{ units.}$$

Alternative Method (using the Vector Product to find the shortest distance between a point and a line):

Convert to a 3D problem. Let $\mathbf{p} = \begin{pmatrix} 0 \\ 2 \\ 0 \end{pmatrix}$. Let A be a known point on line l_2 so that its position vector is $\mathbf{a} = \begin{pmatrix} 0 \\ 8 \\ 0 \end{pmatrix}$. Let $\mathbf{d} = \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix}$. Then

$$h = \frac{|\mathbf{d} \times (\mathbf{p} - \mathbf{a})|}{|\mathbf{d}|} = \frac{\left| \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix} \times \begin{pmatrix} 0 \\ -6 \\ 0 \end{pmatrix} \right|}{\left| \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix} \right|}.$$

$$\begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix} \times \begin{pmatrix} 0 \\ -6 \\ 0 \end{pmatrix} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 1 & 1 & 0 \\ 0 & -6 & 0 \end{vmatrix}$$

$$= (1 \times 0 - 0 \times (-6))\mathbf{i} - (1 \times 0 - 0 \times 0)\mathbf{j} + (1 \times (-6) - 1 \times 0)\mathbf{k} = 0\mathbf{i} + 0\mathbf{j} - 6\mathbf{k}.$$

$$\therefore h = \frac{\left|\begin{pmatrix} 0 \\ 0 \\ -6 \end{pmatrix}\right|}{\left|\begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix}\right|} = \frac{\sqrt{0^2 + 0^2 + (-6)^2}}{\sqrt{1^2 + 1^2 + 0^2}} = \frac{6}{\sqrt{2}} = 3\sqrt{2} \text{ units.}$$

40)

Using a non-Vector method:
Choose a point on l_1: $(0, -20)$. The line passing through $(0, -20)$ which is perpendicular to $y = 5x + 30$ is

$$y + 20 = -\frac{1}{5}(x - 0) \Rightarrow y = -\frac{1}{5}x - 20.$$

$y = -\frac{1}{5}x - 20$ and $y = 5x + 30$ intersect at $(-\frac{125}{13}, -\frac{235}{13})$. Distance between this point and $(0, -20)$ is

$$\sqrt{(0 + \frac{125}{13})^2 + (-20 + \frac{235}{13})^2} = \sqrt{\frac{1250}{13}} = \frac{25\sqrt{26}}{13} \text{ units.}$$

Alternative Method (using Vectors):

Choose a point on l_1, so let A be $(0, -20)$. Let l_2 be $\mathbf{r} = \begin{pmatrix} 0 \\ 30 \end{pmatrix} + \lambda \begin{pmatrix} 1 \\ 5 \end{pmatrix}$. Let P be a point on the line, so $\overrightarrow{OP} = \begin{pmatrix} \lambda \\ 30 + 5\lambda \end{pmatrix}$.

$$\overrightarrow{AP} = \overrightarrow{OP} - \overrightarrow{OA} = \begin{pmatrix} \lambda \\ 30 + 5\lambda \end{pmatrix} - \begin{pmatrix} 0 \\ -20 \end{pmatrix} = \begin{pmatrix} \lambda \\ 50 + 5\lambda \end{pmatrix}.$$

$$\overrightarrow{AP} \cdot \begin{pmatrix} 1 \\ 5 \end{pmatrix} = 0 \Rightarrow \lambda + 5(50 + 5\lambda) = 0 \Rightarrow \lambda + 250 + 25\lambda = 0$$

$$\Rightarrow 26\lambda + 250 = 0 \Rightarrow \lambda = -\frac{125}{13}.$$

So $\overrightarrow{AP} = \begin{pmatrix} -\frac{125}{13} \\ 50 + 5(-\frac{125}{13}) \end{pmatrix} = \begin{pmatrix} -\frac{125}{13} \\ \frac{25}{13} \end{pmatrix}$.

$$|\overrightarrow{AP}| = \sqrt{(-\frac{125}{13})^2 + (\frac{25}{13})^2} = \frac{25\sqrt{26}}{13} \text{ units.}$$

Alternative Method (using the Vector Product to find the shortest distance between a point and a line):

Convert to a 3D problem. Let $\mathbf{p} = \begin{pmatrix} 0 \\ -20 \\ 0 \end{pmatrix}$. Let A be a known point on l_2 so that $\mathbf{a} = \begin{pmatrix} 0 \\ 30 \\ 0 \end{pmatrix}$. Let $\mathbf{d} = \begin{pmatrix} 1 \\ 5 \\ 0 \end{pmatrix}$. Then

$$h = \frac{|\mathbf{d} \times (\mathbf{p} - \mathbf{a})|}{|\mathbf{d}|} = \frac{\left|\begin{pmatrix} 1 \\ 5 \\ 0 \end{pmatrix} \times \begin{pmatrix} 0 \\ -50 \\ 0 \end{pmatrix}\right|}{\left|\begin{pmatrix} 1 \\ 5 \\ 0 \end{pmatrix}\right|}.$$

$$\begin{pmatrix}1\\5\\0\end{pmatrix}\times\begin{pmatrix}0\\-50\\0\end{pmatrix}=\begin{vmatrix}\mathbf{i}&\mathbf{j}&\mathbf{k}\\1&5&0\\0&-50&0\end{vmatrix}$$

$$=(5\times 0-0\times(-50))\mathbf{i}-(1\times 0-0\times 0)\mathbf{j}+(1\times(-50)-5\times 0)\mathbf{k}=0\mathbf{i}+0\mathbf{j}-50\mathbf{k}.$$

$$\therefore h=\frac{\left|\begin{pmatrix}0\\0\\-50\end{pmatrix}\right|}{\left|\begin{pmatrix}1\\5\\0\end{pmatrix}\right|}=\frac{\sqrt{0^2+0^2+(-50)^2}}{\sqrt{1^2+5^2+0^2}}=\frac{50}{\sqrt{26}}=\frac{25\sqrt{26}}{13}\text{ units.}$$

41)

Using a non-Vector method:
Choose a point on l_1: $(0,2)$. The line passing through $(0,2)$ which is perpendicular to $y=-1-10x$ is

$$y-2=\frac{1}{10}(x-0)\Rightarrow y=\frac{1}{10}x+2.$$

$y=\frac{1}{10}x+2$ and $y=-1-10x$ intersect at $(-\frac{30}{101},\frac{199}{101})$. Distance between this point and $(0,2)$ is

$$\sqrt{(0+\frac{30}{101})^2+(2-\frac{199}{101})^2}=\sqrt{\frac{9}{101}}=\frac{3\sqrt{101}}{101}\text{ units.}$$

Alternative Method (using Vectors):

Choose a point on l_1, so let A be $(0,2)$. Let l_2 be $\mathbf{r}=\begin{pmatrix}0\\-1\end{pmatrix}+\lambda\begin{pmatrix}1\\-10\end{pmatrix}$. Let P be a point on the line, so $\overrightarrow{OP}=\begin{pmatrix}\lambda\\-1-10\lambda\end{pmatrix}$.

$$\overrightarrow{AP}=\overrightarrow{OP}-\overrightarrow{OA}=\begin{pmatrix}\lambda\\-1-10\lambda\end{pmatrix}-\begin{pmatrix}0\\2\end{pmatrix}=\begin{pmatrix}\lambda\\-3-10\lambda\end{pmatrix}.$$

$$\overrightarrow{AP}\cdot\begin{pmatrix}1\\-10\end{pmatrix}=0\Rightarrow\lambda+(-3-10\lambda)(-10)=0$$

$$\Rightarrow\lambda+30+100\lambda=0\Rightarrow 101\lambda+30=0\Rightarrow\lambda=-\frac{30}{101}.$$

So $\overrightarrow{AP}=\begin{pmatrix}-\frac{30}{101}\\-3-10(-\frac{30}{101})\end{pmatrix}=\begin{pmatrix}-\frac{30}{101}\\-\frac{3}{101}\end{pmatrix}$.

$$|\overrightarrow{AP}|=\sqrt{(-\frac{30}{101})^2+(-\frac{3}{101})^2}=\sqrt{\frac{9}{101}}=\frac{3\sqrt{101}}{101}\text{ units.}$$

Alternative Method (using the Vector Product to find the shortest distance between a point and a line):

Convert to a 3D problem. Let $\mathbf{p}=\begin{pmatrix}0\\2\\0\end{pmatrix}$. Let A be a known point on l_2 so that $\mathbf{a}=\begin{pmatrix}0\\-1\\0\end{pmatrix}$. Let

$\mathbf{d} = \begin{pmatrix} 1 \\ -10 \\ 0 \end{pmatrix}$. Then

$$h = \frac{|\mathbf{d} \times (\mathbf{p} - \mathbf{a})|}{|\mathbf{d}|} = \frac{\left| \begin{pmatrix} 1 \\ -10 \\ 0 \end{pmatrix} \times \begin{pmatrix} 0 \\ 3 \\ 0 \end{pmatrix} \right|}{\left| \begin{pmatrix} 1 \\ -10 \\ 0 \end{pmatrix} \right|}.$$

$$\begin{pmatrix} 1 \\ -10 \\ 0 \end{pmatrix} \times \begin{pmatrix} 0 \\ 3 \\ 0 \end{pmatrix} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 1 & -10 & 0 \\ 0 & 3 & 0 \end{vmatrix}$$

$$= ((-10) \times 0 - 0 \times 3)\mathbf{i} - (1 \times 0 - 0 \times 0)\mathbf{j} + (1 \times 3 - (-10) \times 0)\mathbf{k} = 0\mathbf{i} + 0\mathbf{j} + 3\mathbf{k}.$$

$$\therefore h = \frac{\left| \begin{pmatrix} 0 \\ 0 \\ 3 \end{pmatrix} \right|}{\left| \begin{pmatrix} 1 \\ -10 \\ 0 \end{pmatrix} \right|} = \frac{\sqrt{0^2 + 0^2 + 3^2}}{\sqrt{1^2 + (-10)^2 + 0^2}} = \frac{3}{\sqrt{101}} = \frac{3\sqrt{101}}{101} \text{ units.}$$

42)

Using a non-Vector method:
l_1 has cartesian equation $y-1 = \frac{1}{3}(x-0) \Rightarrow y = \frac{1}{3}x+1$. l_2 has cartesian equation $y-0 = \frac{1}{3}(x-1) \Rightarrow y = \frac{1}{3}x - \frac{1}{3}$. Choose a point on l_1: $(0,1)$. The line passing through $(0,1)$ which is perpendicular to $y = \frac{1}{3}x - \frac{1}{3}$ is $y - 1 = -3(x-0) \Rightarrow y = -3x + 1$. These lines intersect at $(\frac{2}{5}, -\frac{1}{5})$. Distance between $(\frac{2}{5}, -\frac{1}{5})$ and $(0,1)$ is

$$\sqrt{(0 - \frac{2}{5})^2 + (1 + \frac{1}{5})^2} = \sqrt{\frac{8}{5}} = \frac{2\sqrt{10}}{5} \text{ units.}$$

Alternative Method (using Vectors):

Choose a point on l_1, so let $A = (0,1)$. Let P be a point on l_2, so $\overrightarrow{OP} = \begin{pmatrix} 1 + 3\mu \\ \mu \end{pmatrix}$.

$$\overrightarrow{AP} = \overrightarrow{OP} - \overrightarrow{OA} = \begin{pmatrix} 1 + 3\mu \\ \mu \end{pmatrix} - \begin{pmatrix} 0 \\ 1 \end{pmatrix} = \begin{pmatrix} 1 + 3\mu \\ \mu - 1 \end{pmatrix}.$$

$$\overrightarrow{AP} \cdot \begin{pmatrix} 3 \\ 1 \end{pmatrix} = 0 \Rightarrow (1 + 3\mu)3 + (\mu - 1) = 0$$

$$\Rightarrow 3 + 9\mu + \mu - 1 = 0 \Rightarrow 10\mu + 2 = 0 \Rightarrow \mu = -\frac{1}{5}.$$

So $\overrightarrow{AP} = \begin{pmatrix} 1 + 3(-\frac{1}{5}) \\ -\frac{1}{5} - 1 \end{pmatrix} = \begin{pmatrix} \frac{2}{5} \\ -\frac{6}{5} \end{pmatrix}$ and

$$|\overrightarrow{AP}| = \sqrt{(\frac{2}{5})^2 + (-\frac{6}{5})^2} = \sqrt{\frac{8}{5}} = \frac{2\sqrt{10}}{5} \text{ units.}$$

Alternative Method (using the Vector Product to find the shortest distance between a point and a line):

Convert to a 3D problem. Let $\mathbf{p} = \begin{pmatrix}0\\1\\0\end{pmatrix}$. Let A be a known point on l_2 so that $\mathbf{a} = \begin{pmatrix}1\\0\\0\end{pmatrix}$. Let $\mathbf{d} = \begin{pmatrix}3\\1\\0\end{pmatrix}$. Then

$$h = \frac{|\mathbf{d} \times (\mathbf{p} - \mathbf{a})|}{|\mathbf{d}|} = \frac{\left|\begin{pmatrix}3\\1\\0\end{pmatrix} \times \begin{pmatrix}-1\\1\\0\end{pmatrix}\right|}{\left|\begin{pmatrix}3\\1\\0\end{pmatrix}\right|}.$$

$$\begin{pmatrix}3\\1\\0\end{pmatrix} \times \begin{pmatrix}-1\\1\\0\end{pmatrix} = \begin{vmatrix}\mathbf{i} & \mathbf{j} & \mathbf{k}\\3 & 1 & 0\\-1 & 1 & 0\end{vmatrix} = 0\mathbf{i} + 0\mathbf{j} + 4\mathbf{k}.$$

$$\therefore h = \frac{\left|\begin{pmatrix}0\\0\\4\end{pmatrix}\right|}{\left|\begin{pmatrix}3\\1\\0\end{pmatrix}\right|} = \frac{\sqrt{0^2+0^2+4^2}}{\sqrt{3^2+1^2+0^2}} = \frac{4}{\sqrt{10}} = \frac{2\sqrt{10}}{5} \text{ units.}$$

43)

Using a non-Vector method:
l_1 has cartesian equation $y - 8 = -\frac{5}{4}(x-2) \Rightarrow y = -\frac{5}{4}x + \frac{21}{2}$. l_2 has cartesian equation $y + 4 = -\frac{5}{4}(x+3) \Rightarrow y = -\frac{5}{4}x - \frac{31}{4}$. Choose a point on l_1: $(0, \frac{21}{2})$. The line passing through this point which is perpendicular to $y = -\frac{5}{4}x + \frac{21}{2}$ is $y - \frac{21}{2} = \frac{4}{5}(x-0) \Rightarrow y = \frac{4}{5}x + \frac{21}{2}$. It intersects $y = -\frac{5}{4}x - \frac{31}{4}$ at $(-\frac{365}{41}, \frac{277}{82})$. Distance between this point and $(0, \frac{21}{2})$ is

$$\sqrt{(0 + \frac{365}{41})^2 + (\frac{21}{2} - \frac{277}{82})^2} = \sqrt{\frac{5329}{41}} = \frac{73\sqrt{41}}{41} \text{ units.}$$

Alternative Method (using Vectors):

Choose a point on l_1, so let $A = (2, 8)$. Let P be a point on l_2, so $\overrightarrow{OP} = \begin{pmatrix}-3-4\mu\\-4+5\mu\end{pmatrix}$.

$$\overrightarrow{AP} = \overrightarrow{OP} - \overrightarrow{OA} = \begin{pmatrix}-3-4\mu\\-4+5\mu\end{pmatrix} - \begin{pmatrix}2\\8\end{pmatrix} = \begin{pmatrix}-5-4\mu\\-12+5\mu\end{pmatrix}.$$

$$\overrightarrow{AP} \cdot \begin{pmatrix}-4\\5\end{pmatrix} = 0 \Rightarrow (-5-4\mu)(-4) + (-12+5\mu)5 = 0$$

$$\Rightarrow 20 + 16\mu - 60 + 25\mu = 0 \Rightarrow 41\mu - 40 = 0 \Rightarrow \mu = \frac{40}{41}.$$

So

$$\overrightarrow{AP} = \begin{pmatrix}-5-4(\frac{40}{41})\\-12+5(\frac{40}{41})\end{pmatrix} = \begin{pmatrix}-\frac{365}{41}\\-\frac{292}{41}\end{pmatrix}.$$

$$|\overrightarrow{AP}| = \sqrt{(-\frac{365}{41})^2 + (-\frac{292}{41})^2} = \sqrt{\frac{5329}{41}} = \frac{73\sqrt{41}}{41} \text{ units.}$$

Alternative Method (using the Vector Product to find the shortest distance between a point and a line):

Convert to a 3D problem. Let $\mathbf{p} = \begin{pmatrix} 2 \\ 8 \\ 0 \end{pmatrix}$. Let A be a known point on l_2 so that $\mathbf{a} = \begin{pmatrix} -3 \\ -4 \\ 0 \end{pmatrix}$. Let $\mathbf{d} = \begin{pmatrix} -4 \\ 5 \\ 0 \end{pmatrix}$. Then

$$h = \frac{|\mathbf{d} \times (\mathbf{p} - \mathbf{a})|}{|\mathbf{d}|} = \frac{\left| \begin{pmatrix} -4 \\ 5 \\ 0 \end{pmatrix} \times \begin{pmatrix} 5 \\ 12 \\ 0 \end{pmatrix} \right|}{\left| \begin{pmatrix} -4 \\ 5 \\ 0 \end{pmatrix} \right|}.$$

$$\begin{pmatrix} -4 \\ 5 \\ 0 \end{pmatrix} \times \begin{pmatrix} 5 \\ 12 \\ 0 \end{pmatrix} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ -4 & 5 & 0 \\ 5 & 12 & 0 \end{vmatrix}$$

$$= (5 \times 0 - 0 \times 12)\mathbf{i} - ((-4) \times 0 - 0 \times 5)\mathbf{j} + ((-4) \times 12 - 5 \times 5)\mathbf{k} = 0\mathbf{i} + 0\mathbf{j} - 73\mathbf{k}.$$

$$\therefore h = \frac{\left| \begin{pmatrix} 0 \\ 0 \\ -73 \end{pmatrix} \right|}{\left| \begin{pmatrix} -4 \\ 5 \\ 0 \end{pmatrix} \right|} = \frac{\sqrt{0^2 + 0^2 + (-73)^2}}{\sqrt{(-4)^2 + 5^2 + 0^2}} = \frac{73}{\sqrt{41}} = \frac{73\sqrt{41}}{41} \text{ units.}$$

44)

Using a non-Vector method:
l_1 has cartesian equation $y - k = \frac{5}{2}(x + 4) \Rightarrow y = \frac{5}{2}x + 10 + k$. l_2 has cartesian equation $y - 9 = \frac{5}{2}(x - 8) \Rightarrow y = \frac{5}{2}x - 11$. Choose a point on l_1: $(0, 10 + k)$. The line through this point perpendicular to $y = \frac{5}{2}x - 11$ is

$$y - (10 + k) = -\frac{2}{5}(x - 0) \Rightarrow y = -\frac{2}{5}x + 10 + k.$$

These lines intersect at

$$-\frac{2}{5}x + 10 + k = \frac{5}{2}x - 11 \Rightarrow \frac{29}{10}x = 21 + k \Rightarrow x = \frac{210}{29} + \frac{10}{29}k.$$

$$\therefore y = \frac{5}{2}(\frac{210}{29} + \frac{10}{29}k) - 11 = \frac{25}{29}k + \frac{206}{29}.$$

The lines intersect at $\left(\frac{210}{29} + \frac{10}{29}k, \frac{25}{29}k + \frac{206}{29}\right)$. The distance to $(0, 10 + k)$ is $4\sqrt{29}$:

$$\sqrt{(0 - (\frac{210}{29} + \frac{10}{29}k))^2 + ((10 + k) - (\frac{25}{29}k + \frac{206}{29}))^2} = 4\sqrt{29}$$

$$\Rightarrow (\frac{210}{29} + \frac{10}{29}k)^2 + (\frac{4}{29}k + \frac{84}{29})^2 = 464$$

$$\Rightarrow (\frac{1}{29})^2(210 + 10k)^2 + (\frac{1}{29})^2(4k + 84)^2 = 464$$

$$\Rightarrow (210 + 10k)^2 + (4k + 84)^2 = 390224$$

$$\Rightarrow 100k^2 + 4200k + 44100 + 16k^2 + 672k + 7056 = 390224$$

$$\Rightarrow 116k^2 + 4872k - 339068 = 0 \Rightarrow k^2 + 42k - 2923 = 0$$
$$\Rightarrow (k+79)(k-37) = 0 \Rightarrow k = -79 \text{ or } k = 37.$$

Alternative Method (using Vectors):

Choose a point on l_1, so let $A = (-4, k)$. Let P be a point on l_2, so $\overrightarrow{OP} = \begin{pmatrix} 8+2\mu \\ 9+5\mu \end{pmatrix}$.

$$\overrightarrow{AP} = \overrightarrow{OP} - \overrightarrow{OA} = \begin{pmatrix} 8+2\mu \\ 9+5\mu \end{pmatrix} - \begin{pmatrix} -4 \\ k \end{pmatrix} = \begin{pmatrix} 12+2\mu \\ 9-k+5\mu \end{pmatrix}.$$

$$\overrightarrow{AP} \cdot \begin{pmatrix} 2 \\ 5 \end{pmatrix} = 0 \Rightarrow (12+2\mu)2 + (9-k+5\mu)5 = 0$$

$$\Rightarrow 24 + 4\mu + 45 - 5k + 25\mu = 0 \Rightarrow 29\mu + 69 - 5k = 0 \Rightarrow \mu = \frac{5}{29}k - \frac{69}{29}.$$

So

$$\overrightarrow{AP} = \begin{pmatrix} 12 + 2(\frac{5}{29}k - \frac{69}{29}) \\ 9 - k + 5(\frac{5}{29}k - \frac{69}{29}) \end{pmatrix} = \begin{pmatrix} \frac{10}{29}k + \frac{210}{29} \\ -\frac{4}{29}k - \frac{84}{29} \end{pmatrix}.$$

$$|\overrightarrow{AP}| = 4\sqrt{29} \Rightarrow \sqrt{(\frac{10}{29}k + \frac{210}{29})^2 + (-\frac{4}{29}k - \frac{84}{29})^2} = 4\sqrt{29}$$
$$\Rightarrow (\frac{1}{29})^2(10k+210)^2 + (\frac{1}{29})^2(4k+84)^2 = 464$$
$$\Rightarrow (10k+210)^2 + (4k+84)^2 = 390224$$
$$\Rightarrow 100k^2 + 4200k + 44100 + 16k^2 + 672k + 7056 = 390224$$
$$\Rightarrow 116k^2 + 4872k - 339068 = 0 \Rightarrow k^2 + 42k - 2923 = 0$$
$$\Rightarrow (k+79)(k-37) = 0 \Rightarrow k = -79 \text{ or } k = 37.$$

Alternative Method (using the Vector Product to find the shortest distance between a point and a line):

Convert to a 3D problem. Let $\mathbf{p} = \begin{pmatrix} -4 \\ k \\ 0 \end{pmatrix}$. Let A be a known point on l_2 so that $\mathbf{a} = \begin{pmatrix} 8 \\ 9 \\ 0 \end{pmatrix}$. Let $\mathbf{d} = \begin{pmatrix} 2 \\ 5 \\ 0 \end{pmatrix}$. Then

$$h = \frac{|\mathbf{d} \times (\mathbf{p} - \mathbf{a})|}{|\mathbf{d}|} = 4\sqrt{29} \Rightarrow \frac{\left| \begin{pmatrix} 2 \\ 5 \\ 0 \end{pmatrix} \times \begin{pmatrix} -12 \\ k-9 \\ 0 \end{pmatrix} \right|}{\left| \begin{pmatrix} 2 \\ 5 \\ 0 \end{pmatrix} \right|} = 4\sqrt{29}.$$

$$\begin{pmatrix} 2 \\ 5 \\ 0 \end{pmatrix} \times \begin{pmatrix} -12 \\ k-9 \\ 0 \end{pmatrix} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 2 & 5 & 0 \\ -12 & k-9 & 0 \end{vmatrix}$$

$$= (5 \times 0 - 0 \times (k-9))\mathbf{i} - (2 \times 0 - 0 \times (-12))\mathbf{j} + (2(k-9) - 5(-12))\mathbf{k}$$
$$= 0\mathbf{i} + 0\mathbf{j} + (2k+42)\mathbf{k}.$$

$$\therefore h = \frac{\left|\begin{pmatrix} 0 \\ 0 \\ 2k+42 \end{pmatrix}\right|}{\left|\begin{pmatrix} 2 \\ 5 \\ 0 \end{pmatrix}\right|} = 4\sqrt{29} \Rightarrow \frac{\sqrt{(2k+42)^2}}{\sqrt{29}} = 4\sqrt{29}$$

$$\Rightarrow \sqrt{(2k+42)^2} = 116 \Rightarrow 2k+42 = \pm 116.$$

$$2k+42 = 116 \Rightarrow k = 37, \qquad 2k+42 = -116 \Rightarrow k = -79.$$

So $k = -79$ or $k = 37$.